

Teacher Guide: Week 1

Logic Is Not Magic: What Logic Can and Cannot Do

10th Grade CUE Test Run Based on Richard Whately's *Elements of Logic*

Week 1 Overview

This first week introduces Richard Whately's basic account of logic. Students should leave the week understanding that logic is not a machine that automatically discovers truth. Logic studies the structure of reasoning and helps expose bad inferences. This establishes the foundation for later weeks on terms, propositions, arguments, fallacies, ambiguity, induction, and verbal versus real disputes.

Essential Question

What can logic actually do for a thinker, and what should we not expect it to do?

Student-Facing Materials

- *Logic Is Not Magic: What Logic Can and Cannot Do* — student reading handout.
- *Week 1 Argument Lab: What Follows From What?* — practice handout and window-note task.

Learning Goals

Students will be able to:

1. Define logic as the study of reasoning and argument structure.
2. Distinguish what logic can do from what it cannot do.
3. Identify conclusion, stated reason, and hidden assumption in a simple argument.
4. Explain why confidence, popularity, age, or emotion do not by themselves prove a claim.
5. Write a short claim paragraph using an analogy for logic.

Key Vocabulary

logic, reasoning, argument, premise, conclusion, deduction, fallacy, science, art, panacea, hidden assumption

Weekly Lesson Sequence

Day 1: Opening Question and First Read

- Begin with the question: *If someone is good at arguing, does that mean he is good at finding truth?*
- Have students answer briefly in writing.
- Introduce Whately as a nineteenth-century thinker trying to rescue logic from both overpraise and contempt.
- Read the introduction and first half of the student reading aloud or in pairs.
- Stop after the analogy paragraph. Ask: What does Whately think logic is being unfairly expected to do?

Day 2: Finish Reading and Annotate Whately's Claim

- Finish the reading.
- Students mark three places: one sentence showing what logic can do, one showing what it cannot do, and one example that separates conclusion, stated reason, hidden assumption, and test.
- Complete Reading Focus questions 1–6.

Day 3: Argument Skeleton Practice

- Introduce the pattern: conclusion, stated reason, hidden assumption, test.
- Work through the example on the Argument Lab handout.
- Students complete Part I individually, then compare answers with a partner.
- Emphasize that Whately's basic question is: *What follows from what?*

Day 4: Window Notes and Discussion

- Students complete the window-note task: what logic can do versus what it cannot do.
- Require a visual component and a written explanation.
- Discuss several student examples. Push students to separate proof from confidence, popularity, and emotion.

Day 5: Exit Claim Paragraph

- Students write the exit claim paragraph: *Logic is most like* _____ *because* _____.
- Require one reference to Whately's view and one example from school, literature, public speech, or everyday life.
- Collect the student reading and argument lab as the week's work product.

Suggested Board Notes

- Logic does not discover every truth.
- Logic studies whether a conclusion follows from premises.
- Bad facts can still produce bad conclusions even when the reasoning looks neat.
- Strong feelings, popularity, age, confidence, and repetition are not the same as proof.
- The first logical question: What follows from what?

Reading Focus Answer Guide

1. Main claim: Logic is the science and art of reasoning; it studies argument structure and helps guard against bad deductions.
2. Can do: examine whether conclusions follow; expose skipped steps, changed terms, hidden assumptions, or bad inferences.
3. Cannot do: supply facts by itself; make foolish people wise; discover every truth; replace observation, experience, memory, imagination, or moral judgment.
4. Four steps: find the conclusion, find the stated reason, identify the hidden assumption, and test whether the conclusion really follows.
5. Answers will vary. Strong responses should use one reading example and correctly name its conclusion, reason, assumption, and test. For example, the Macbeth argument concludes that Macbeth is not responsible; its stated reason is the witches' prediction; its hidden assumption is that prediction removes responsibility; the conclusion does not necessarily follow.
6. People blamed logic because some defenders promised too much for it, so critics rejected it when it failed to perform impossible tasks.

Argument Lab Answer Guide

Argument 1: Macbeth

- Conclusion: Macbeth cannot be responsible for what he does afterward.

- Stated reason: The witches predicted his future.
- Hidden assumption: If an event is predicted, the person involved is not responsible for his choices.
- Test: The conclusion does not necessarily follow. Prediction, influence, temptation, and responsibility are different ideas.

Argument 2: Confidence and Proof

- Conclusion: Confidence is not the same thing as proof.
- Stated reason: A person can speak confidently and still be wrong.
- Hidden assumption: If two things can come apart, they are not identical.
- Test: This is a strong argument if students understand that confidence can accompany both truth and error.

Argument 3: Old Ideas

- Conclusion: The idea has nothing to teach modern students.
- Stated reason: The idea is old.
- Hidden assumption: Old ideas cannot teach modern students.
- Test: The conclusion does not follow. Some old ideas are false or obsolete, but age alone does not prove uselessness.

Assessment

Use the exit paragraph as a formative assessment. A strong paragraph should:

- make a clear analogy for logic;
- explain both usefulness and limits;
- refer to reasoning, argument, or proof;
- include a concrete example;
- avoid treating logic as mere debate tricks.

Next Week Preview

Week 2 should move from reasoning in general to **terms**: the words that carry arguments. The next guiding question should be: *Why do arguments become unstable when key words are unclear?*